S.I.R.E.

SIMULATOR FOR INCIDENT RESPONSE EXERCISES

Real-time, scenario-driven emergency training — in any browser

Team Nibble
NSCC · PROG 3300
Integrated Projects for Programming
Instructor: Alfred Parkes

WHAT IS S.I.R.E.?

S.I.R.E. is a **web-based incident response training platform** that lets instructors run live, scenario-driven emergency drills for groups of any size — no specialised hardware or software installation required.

Trainees join from any device using a session code, receive real-time escalating events, log their actions, and receive instant feedback — all synchronised through a persistent WebSocket connection.

HOW IT WORKS

1. **Instructor creates a session** — picks a scenario and shares the 6-character session code.
2. **Trainees join instantly** — enter the code and display name in any browser; no account needed.
3. **Scenario runs automatically** — timed events escalate and broadcast to every participant in real time.
4. **Trainees log actions** — responses and rationale are captured and visible to the instructor live.
5. **Instructor injects events** — custom escalations or curve-balls can be broadcast at any moment.

KEY FEATURES

Instructor

- Create & manage sessions
- Select from 8 scenarios
- Start / pause / end drills
- Inject custom events live
- Monitor trainee roster
- View live event feed

Trainee

- Join with code only
- See real-time events
- Log actions + rationale
- Track full event history
- Colour-coded severity
- Works on any device

8 TRAINING SCENARIOS

Active Threat — Lockdown & law-enforcement coord.	**Fire Emergency** — Alarm, evacuation, fire dept. handover
Flood — Water intrusion & system protection	**Cyber Attack** — Ransomware, isolation & recovery
Power Outage — UPS/generator failover	**Severe Weather** — Shelter-in-place & facility lockdown
Medical Emergency — Cardiac event, AED & EMS coord.	**Hazmat Spill** — Containment, PPE & decontamination

SYSTEM ARCHITECTURE

Instructor Dashboard ⇄ **Trainee Interface**

↕ HTTPS / Socket.IO (WSS)

REST API — Express · Node.js

Socket.IO — Real-time events

↕ In-process

Session Service & Scenario Engine — In-memory session store · JSON scenario definitions · timed event dispatcher

↑ Deployed to

Render Cloud — sire-api (Node) & sire-web (Static)

Sessions isolated via Socket.IO rooms · Events auto-broadcast on timer · API-key & ticket-ID auth

TECHNOLOGY STACK

FRONTEND

React 19 · Vite 7 · MUI v7 · React Router v7 · ESM / JSX

BACKEND

Node.js 25+ · Express · ESM (.mjs)

REAL-TIME

Socket.IO · WebSocket (WSS) · Namespaced Rooms

DEVOPS & DEPLOYMENT

Render (Cloud) · Docker · Railway · GitHub

KEY DESIGN DECISIONS

- **Socket.IO over plain WebSocket** — built-in rooms, reconnection & namespacing provide seamless multi-session isolation.
- **In-memory session store** — zero external dependencies; ideal for short-lived academic drills with no persistence overhead.
- **JSON scenario definitions** — instructors and non-developers can author or modify scenarios without touching code.
- **Vite + React frontend** — instant HMR in dev; optimised static bundle at build time, served from a global CDN.
- **API-key + ticket-ID auth** — lightweight, auditable security model suited to academic and demo deployments.

THE TEAM

Tom Burchell
Backend & Real-Time Lead
Socket.IO · scenario engine · server architecture · CI/CD · cloud deployment

Leon Wasiliew
Frontend & UI/UX Lead
React UI · layouts · real-time state · responsive design

John Hay
Script Master / QA
Scenario writing · integration testing · accessibility review

Kael Payette
Project Manager
Coordination · documentation · Git workflow · standards